

WEEK 1 ASSIGNMENT

AI/ML Foundations & Data Science Exploration

Team: VERSATILE

GETTING STARTED

Group Details & Task Split

- | | | |
|---|--------------------|-----------------------------|
| 1 | Nanda | Task 1 - AI Tools We Use |
| 2 | Sai Tejaswi | Task 2 - Python in Industry |
| 3 | Anuya | Task 3 - Data Wrangling |
| 4 | Keerthana | Task 4 - Data Storytelling |
| 5 | Mohit Kumar Sharma | Task 5 - GitHub Setup |

Each member researched and explained their task to the group during our discussion call.

TASK 1

AI Tools We Already Use

Most of us use AI tools every day without even thinking of them as "AI". Here are five tools from our own phones and laptops, and what's actually happening behind the screen.





Ollama

Ollama lets us run AI language models right on our own laptop, no internet needed and nothing sent to a server. We use it for quick coding doubts when campus wifi is slow. Since everything runs locally, the chats stay private and it costs nothing to use. The best part is speed - answers come instantly, no waiting for a website to respond.



Grammarly

Grammarly checks our writing as we type, whether it's a college report, an email to a professor, or a LinkedIn post. It runs as a browser extension and flags spelling, grammar, and tone issues. We use it almost daily for assignments. It also explains why something is wrong, so our own writing slowly gets better, saving a lot of last-minute editing time.



Google Lens

Google Lens turns the phone camera into a search bar. Point it at a plant, a math problem, or a product, and it tells you what it is. We mostly use it to translate text inside images, like a notice board or a textbook page, and to identify objects fast. Searching becomes visual instead of typing - useful when you don't even know the right words to search.



Google Translate

Google Translate helps us read research papers, reviews, and messages written in languages we don't know. We use it for text, speech, and even camera input across 100+ languages. It studies patterns across huge amounts of translated text to sound natural rather than word-by-word. This removes the language barrier completely during group projects with people from different states.



Blackbox AI

Blackbox AI is a coding assistant we use inside the browser and code editor. It auto-completes code, explains confusing snippets, and converts code between languages. As CSE students still learning, this helps a lot while debugging or reading someone else's code. Instead of getting stuck for an hour, we get an explanation right away and keep moving on the assignment.

TASK 2

Python in Industry

We picked two companies from very different worlds - professional services and global banking - to see why both still rely heavily on Python.





EY (Ernst & Young)

EY is one of the world's largest professional services firms, working in audit, tax, and business consulting for companies worldwide. A huge part of their job involves analysing large volumes of financial data, and Python makes that process faster and far more accurate than doing it by hand. EY uses Python to automate repetitive audit checks, scan financial statements for unusual patterns that might point to fraud, and build internal dashboards for clients. Its data libraries also help EY's analytics teams build risk models quickly. Since EY handles confidential financial data at scale, Python's reliability is exactly what they need.



JPMorgan Chase

JPMorgan is one of the largest banks in the world, handling trading, investment banking, and asset management for clients globally. Trading decisions depend on processing massive amounts of market data within seconds, and Python's speed for prototyping fits perfectly here. JPMorgan even built and open-sourced its own Python framework, Athena, used internally for risk analysis and trading strategies. Python builds pricing models, flags unusual transactions, and automates reports that once took analysts hours. Since markets change every second, a language built for fast testing gives JPMorgan a real edge.

TASK 3



Data Wrangling: A Movie Dataset

We picked an IMDb movie dataset listing thousands of films with their budget, revenue, rating, runtime and release year. Looking closely, four problems showed up again and again:



Missing Values

Many older films have no budget or revenue listed - this data simply wasn't tracked publicly before the 2000s.



Duplicate Records

A handful of movies appear twice, usually added once under the original title and again under a re-release title.



Incorrect Values

Some rows show a runtime of 0 minutes or a release year like 2106 - clearly typing mistakes during entry.



Outliers

One or two movies list budgets in the billions - far beyond any real film budget, so likely entry errors.

Why This Matters for Machine Learning

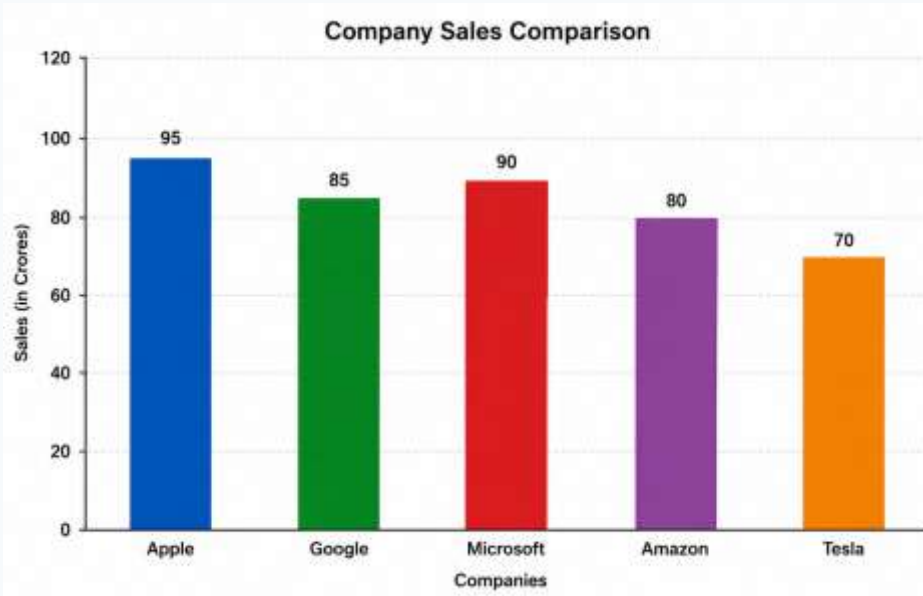
These issues matter a lot once this data trains a machine learning model. Missing values confuse most algorithms - they can't handle empty cells, and dropping every such row means losing useful data. Duplicate records quietly push the model to think certain patterns are more common than they really are, which leads to overfitting. Wrong values, like a runtime of 0, teach the model a completely false relationship between features and the outcome. Outliers are the trickiest part - a single billion-dollar budget can pull the average so far off that the model starts treating normal-budget films as the unusual ones.

So before any analysis or prediction work begins, cleaning this data isn't optional - it's the real first step of the project.

TASK 4



Storytelling with Data: BUSINESS



- What it shows: The graph compares the sales revenue of different companies. It helps us understand which company has achieved higher sales..
- Insight: Apple has the highest sales among all the companies, indicating strong market performance.
- Decision: Companies with lower sales can improve their marketing strategies and introduce innovative products to increase revenue.

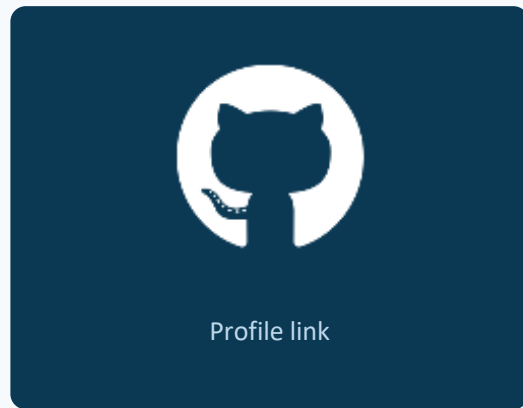
TASK 5



GitHub Profile Setup

GitHub is where developers store and share code, built on Git for tracking every change made to a project. As CSE students, it's where we keep our work organised and visible to others - recruiters use it to judge real project experience, not just marks.

- Repositories for our portfolio site, ML projects, and college assignments
- Every change tracked through commits - easy to undo mistakes
- README files that explain what each project does and how to run it
- Branches to try new features without breaking the main project
- A growing, public record of what we've actually built



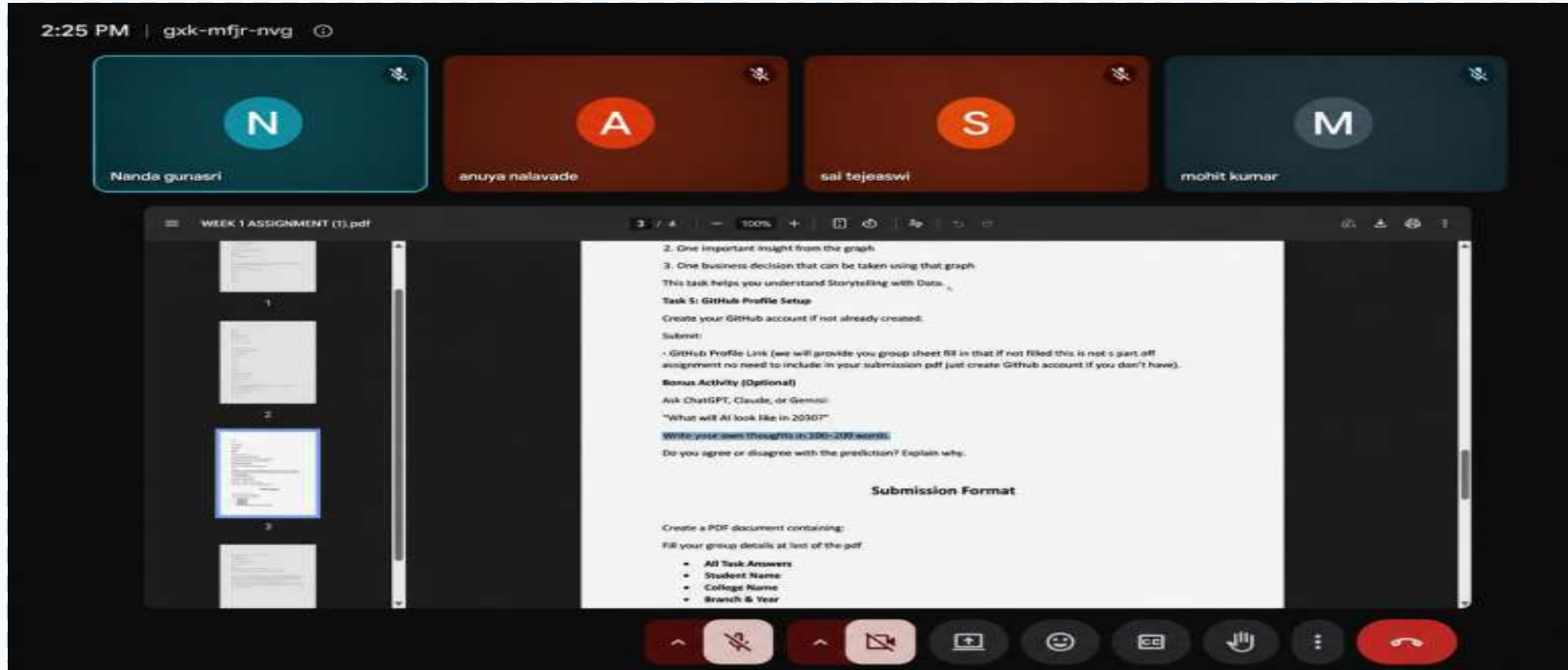
GROUP DISCUSSION

Google Meet Screenshot



GROUP DISCUSSION

Google Meet Screenshot (cont.)





Thank You

Team Versatile